

Radiative processes in astrophysics (First part of PHY-404.1)

March 17th, 2026

Procedure for the submission of the assignment:

- Assignment is advised to be submitted as a typed report in Latex in a PDF format.
- For each question in the assignment sheet, make a section.
- A GitHub link to the codes should be mentioned in each section.
- In the acknowledgment, mention if anyone else has helped you in solving a problem. Taking help from someone is solving a problem without acknowledging it is not allowed.
- Cite any numerical packages used in the analysis.
- Only use codes developed by you for the solving the problems.
- Deadline for the submission of the assignment: **Before the end of March 27th, 2026.**

1 Assignment-II

1. Consider a spherical system of radius r_0 having particles with constant number density in thermal distribution in which the temperature varies with radius of the system r as (i) $T(r) = 10^4 K \times (r/r_0)$ and (ii) $T(r) = 10^6 K \times \exp(-r/r_0)$. Calculate and plot the observed intensity for particles undergoing bremsstrahlung emission and absorption as a function of radius from the center up to r_0 for frequency range $10^6 - 10^{16}$ Hz. [You can consider charge and mass of the particles as your preference. But give these details in your answer. The Gaunt factor can be kept as a factor without putting any numerical value.]
2. A) Calculate numerically and plot the electric field and magnetic field for a relativistically moving electron in the presence of a proton with velocity (a) $0.1c$, (ii) $0.999c$. (B) Calculate the bremsstrahlung emission spectra for both the cases. [The Gaunt factor can be kept as a factor without putting any numerical value].

3. Suppose a system of length $L=10^{17}$ cm is having constant and equal density of electrons and protons n_e and n_p respectively which follows a Maxwellian distribution at a temperature T . For cases with n_e and n_p varying from 10^9 per cm^3 to 10^{20} per cm^3 , show the bremsstrahlung intensity spectra is the presence of both absorption and emission as a function of frequency. Discuss in which limit the spectra becomes close to blackbody and show the relative difference with respect to a blackbody at the same temperature T .
4. Calculate (A) the total power emitted P_{sync} in synchrotron emission and (B) the cooling timescale by an electron in (i) magnetic field 10^{-1} Gauss and velocity $v = 0.99c$, (ii) magnetic field 10^8 Gauss and velocity $v = 0.999c$, (iii) magnetic field 10^8 Gauss and velocity $v = 0.1c$. (C) If we assume a condition to obtain the synchrotron emission that in one orbit, the energy of the particle is not changed significantly, then calculate the condition on the strength of the magnetic field for this assumption to be valid.
5. Show the key differences between the cyclotron emission spectra and synchrotron emission spectra due to an electron moving in the presence of a magnetic field of 10^8 Gauss with velocity $v = 0.01c$ and $v = 0.999c$.
6. Compare the emission spectra properties of bremsstrahlung and synchrotron emission and show the key physical reasons behind the differences in the spectra.

Optional As suggested in the class, if you have read any research paper(s) or have done any literature survey to find out astrophysical applications of bremsstrahlung and synchrotron mechanisms, you can list down the reference details of those papers in your assignment.